



Amusement Research Project

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Project Idea

Our project aims to develop a standalone desktop application capable of processing audio-video input to detect a person's laughter and overall sense of amusement. The system interprets facial expressions and vocal cues using scientifically grounded methods derived from affective computing and facial expression theory.



Research Purpose

- Find patterns in the sense of humor of people aged 18-35
- Better understand the differences between male and female humor
- Find the all around greatest contribution factor to people's laughter
- Qualify human positive emotions in a controlled environment

Research Questions

Q1: What category of videos do people find the funniest?

1. Funny animals
2. Human fails
3. Trending Memes/Internet Humor
4. Pranks/Hidden Camera clips
5. Reaction Humor
6. AI-powered clips
7. Culture-Related Humor

Are there any differences between male and female participants?

Q2: How does the presence of others influence the amusement intensity?

Scientific Basis

Facial Action Coding System (FACS)

FACS is a scientific method used to describe facial expressions by breaking them into small muscle movements called Action Units (AUs). Instead of guessing emotions, FACS lets us identify them based on measurable changes in the face.

Key AUs for smiling and laughing:

- **AU12** – Lip Corner Puller: The mouth corners lift upward. This is the main indicator of a smile.
- **AU6** – Cheek Raiser: The cheeks rise and the eyes get slightly narrower. When AU6 appears together with AU12, it usually means a genuine smile or real amusement.
- **AU25 / AU26** – Mouth Opening: AU25 means the lips part; AU26 means the jaw drops further. These are common when someone is laughing, especially during stronger laughter.

Affective Computing Principles

Affective computing is the field that studies how computers can understand human emotions. It says emotions can be detected from:

- Facial expressions
- Voice sounds

Our app follows these ideas by combining facial and audio cues. Using more than one type of signal makes our amusement detection more accurate.

Quantitative Amusement Measurement

Our system turns smiles and laughter into numbers. We calculate values like:

- Smile strength
- How open the mouth is

These numbers create an "Amusement Score." This allows us to compare amusement between people, video types, and situations in a fair and scientific way.

Illustration - FACS

This chart shows the full range of Facial Action Units (AUs) defined by the Facial Action Coding System (FACS). Each AU corresponds to a specific facial muscle movement, such as raising the cheeks, pulling the lip corners, or opening the jaw. By breaking expressions into these components, FACS allows us to analyze amusement objectively. In our project, we focus mainly on the AUs most linked to smiling and laughter—particularly AU6, AU12, and AU25/26—while understanding that these reactions are part of a broader system of coordinated facial actions.

AU 1	AU 2	AU 4	AU 5	AU 6	AU 7
					
Inner Brow Raiser	Outer Brow Raiser	Brow Lowerer	Upper Lid Raiser	Cheek Raiser	Lid Tightener
*AU 41	*AU 42	*AU 43	AU 44	AU 45	AU 46
					
Lid Droop	Slit	Eyes Closed	Squint	Blink	Wink
Lower Face Action Units					
AU 9	AU 10	AU 11	AU 12	AU 13	AU 14
					
Nose Wrinkler	Upper Lip Raiser	Nasolabial Deepener	Lip Corner Puller	Cheek Puffer	Dimpler
AU 15	AU 16	AU 17	AU 18	AU 20	AU 22
					
Lip Corner Depressor	Lower Lip Depressor	Chin Raiser	Lip Pucker	Lip Stretcher	Lip Funneler
AU 23	AU 24	*AU 25	*AU 26	*AU 27	AU 28
					
Lip	Lip	Lips	Jaw	Mouth	Lip

Technical Explanation

Models Used in Our System

MediaPipe Face Mesh

- Detects 468 facial landmarks in real time
- Allows us to measure smile and laughter indicators such as:
- Mouth openness (AU25 / AU26)
- Mouth corner movement (AU12)
- Cheek raising (AU6)

Audio Activity Detection (WebRTC VAD)

- Detects whether voice-like sound is present
- Helps identify laughter by checking if:
- The mouth is open and
- There is an audio burst

How We Use Them Together: If the mouth is open (Face Mesh)
+ audio activity detected (VAD) → likely laughter

Action Unit Computation Formulas

We compute Action Units (AUs) using Euclidean distances between facial landmarks.
All AUs are normalized and clipped so that $AU \in [0,1]$.

Distance

Let $d(A, B)$ = Euclidean distance between points A and B

Normalization Factors

Horizontal AUs (smile / cheek movement)

Let $Norm_H = d(\text{Left Eye Inner Corner}, \text{Right Eye Inner Corner})$

Vertical AUs (mouth opening / jaw drop)

Let $Norm_V = d(\text{Nose Tip}, \text{Chin Tip})$

Activation Function

$$AU = \min(\max(AU, 0), 1)$$

AU Formulas (Using Facial Feature Names)

AU12 — Lip Corner Puller (smile width)

$$AU_{12} = \frac{d(\text{Left Mouth Corner}, \text{Right Mouth Corner})}{Norm_H}$$

AU6 — Cheek Raiser (eye–cheek compression)

Left cheek:

$$AU_{6_L} = 1 - \frac{d(\text{Left Cheek}, \text{Left Lower Eyelid})}{Norm_H}$$

Right cheek:

$$AU_{6_R} = 1 - \frac{d(\text{Right Cheek}, \text{Right Lower Eyelid})}{Norm_H}$$

Final AU6:

$$AU_6 = \frac{(AU_{6_L} + AU_{6_R})}{2}$$

AU25 — Lips Part (lip separation)

$$AU_{25} = \frac{d(\text{Upper Lip Center}, \text{Lower Lip Center})}{Norm_V}$$

AU26 — Jaw Drop

$$AU_{26} = \frac{d(\text{Lower Lip Center}, \text{Chin Tip})}{Norm_V}$$

Audio Activity (WebRTC VAD)

$$\text{AudioActivity} = \begin{cases} 1 & \text{if sound is detected} \\ 0 & \text{otherwise} \end{cases}$$

Scoring Methodology

A. Smile Score

We want to capture how wide the smile is + how genuine it is.

$$SmileScore = 0.6 \cdot AU12 + 0.4 \cdot AU6$$

Where AU12 = mouth corners going up and AU6 = cheek raising (genuine smile)

B. Laughter Score

Laughter includes smiling, cheeks raising, mouth opening, and audio bursts.

$$LaughterScore = 0.3 \cdot AU12 + 0.3 \cdot AU6 + 0.2 \cdot AU25 + 0.2 \cdot AudioActivity$$

Where AU25/AU26 = mouth openness and AudioActivity = 1 when VAD detects sound, 0 otherwise

C. Amusement Score

We give slightly more weight to laughter because it signals stronger amusement:

$$AmusementScore = 0.6 \cdot LaughterScore + 0.4 \cdot SmileScore$$

Result is always between 0 and 1.

Video Analysis Metrics

A. Average scores over the whole video

$$AvgSmile = \frac{1}{T} \sum_{t=1}^T SmileScore(t)$$

$$AvgLaughter = \frac{1}{T} \sum_{t=1}^T LaughterScore(t)$$

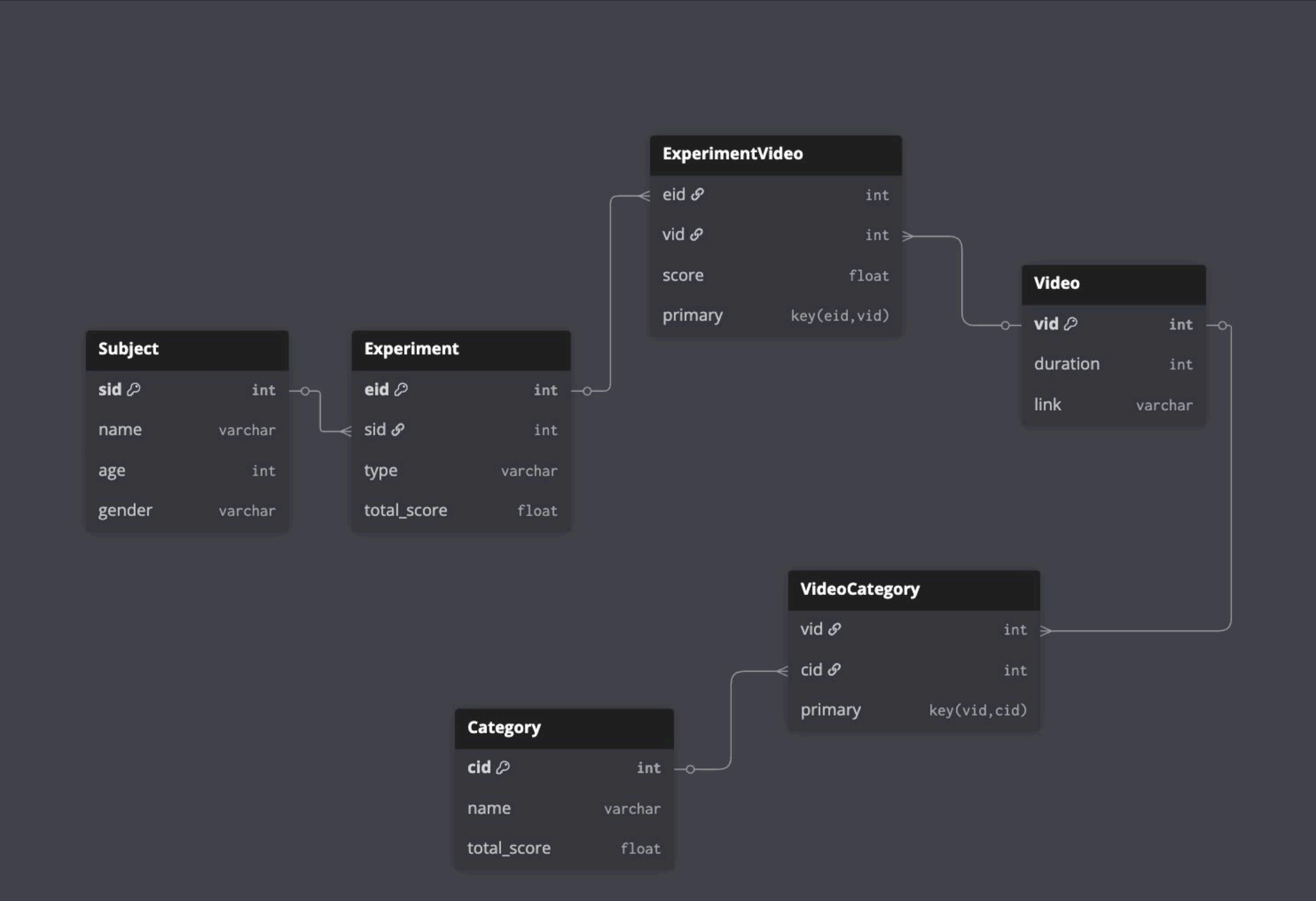
$$AvgAmusement = \frac{1}{T} \sum_{t=1}^T AmusementScore(t)$$

Where T is the number of frames (or time steps) in the video.

Final Analysis

- Identify which video categories produce the strongest amusement
- Compare male vs. female amusement patterns
- Measure how social context (alone vs. with others) affects laughter

Database Design and use of persistency



Each experiment session will have a fixed duration of **7 minutes**, during which a randomised sequence of humorous videos will be played. We include a wide variety of clips across all previously defined categories, and each video’s link and duration are stored in the database. Throughout the 7-minute session, the participant can see the elapsed time in real-time while our system continuously analyzes their facial expressions and audio input to detect laughter and amusement.

Because of our database design, every experiment is linked to the exact set of videos shown during that session, and for each video we store the amusement score produced by the participant. This allows us to later determine which clips triggered the strongest reactions and how different subjects responded to different categories. The database also stores participant information (age, gender, name) and general experiment details (type: single or group, total score), enabling us to compare reactions across individuals and conditions. In addition, as some videos could belong to multiple categories, we established a table named 'Category' and a many-to-many relationship between Category and Videos.

Making use of the Youtube video links

For each experiment, our application will dynamically build a YouTube “playlist” in real time using random video links stored in the database. Before the session starts, the system selects a random combination of clips from all categories whose total duration is as close as possible to the 7-minute limit. These video IDs are then passed to the YouTube player as a sequence, using YouTube’s ability to automatically chain multiple videos together without us having to create a manual playlist on the platform. During playback, the script shows a live timer, automatically loads the next video when one ends, and stops the player exactly when the 7 minutes are over. Throughout this continuous playlist, our program records the user’s reactions (face and audio) while keeping a precise link between each played video and its corresponding amusement score in the database.

Example: https://www.youtube.com/watch_videos?video_ids=VIDEOID1,VIDEOID2,VIDEOID3 will create a playlist of 3 videos.

Data Collection Process

01

Participant Setup

We aim to experiment on 50 different subjects using the webcam and microphone to record video and audio in real time.

03

Audio Analysis

WebRTC VAD checks if voice-like sound is detected in the audio.

02

Facial Landmark Extraction

MediaPipe Face Mesh extracts facial landmarks from each video frame, measuring smile and laughter indicators such as mouth openness (AU25/AU26), mouth corner movement (AU12), and cheek raising (AU6).

04

Data Storage

All extracted values (AUs, sound activity, timestamps) are saved into a local dataset for analysis.



Experiment Conditions

The experiment will be conducted in a quiet, controlled room to minimise background noise, as our audio module is sensitive and requires silence for accurate detection. Participants will be instructed to avoid talking, chewing, or making deliberate facial movements during the clips, ensuring that any mouth activity can be interpreted as genuine amusement. We will use a DJI OSMO Pocket 3 camera as a web cam which has the capability to film in resolutions up to 4K, and also has FaceTracking capabilities, in order to ensure maximum quality analysis.

Participant Structure

Approximately 50 participants will be recruited and shown a series of humorous video clips while their facial and vocal reactions are recorded. To investigate how social context influences amusement, a subgroup of 10 participants will complete two sessions: one individually, and one in a shared viewing setting with all 10 participants present in the same room.

Real World Impact

- Providing useful insights for content creators in order to create more personalized products and experiences
- The ability to turn human reactions into objective, measurable data
- Improving the marketing choices for advertising



Limitations

Poor Lighting Conditions

Facial expression recognition can be significantly hampered by suboptimal lighting, potentially leading to inaccurate detection of AUs and affecting the overall amusement score.

Small Sample Size

Our experiment involves approximately 50 participants, which, while offering initial insights, may not be sufficiently large or diverse to generalize findings across broader populations or demographics.

Limited Audio Classification

The WebRTC VAD primarily detects the presence of voice activity and cannot distinguish between laughter and other vocalizations. This means some audio bursts might be misclassified, affecting the precision of laughter scoring.

Future Work

- Expand the dataset with more participants and more diverse age groups
- Replace WebRTC VAD with a real laughter classification model

Conclusion

We propose a scientifically grounded system for detecting amusement from audio-video input. This system utilizes precise facial landmark analysis and lightweight audio cues to transform subjective human reactions into objective, measurable data.

Our experiment will investigate two core questions: determining which video categories elicit the strongest amusement, and analyzing how the presence of others influences amusement intensity. These insights are crucial for understanding the nuances of human emotional responses.

This robust methodology and controlled experimental conditions will allow us to objectively measure and compare laughter and smile responses across diverse participants and various viewing settings, providing a new benchmark for amusement research.



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Thank you!